COMP3670/6670
Introduction to Machine Learning
Week 6a: Matrices, Linear Maps & Systems

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Goals of this lecture

- Organise datasets and image batches using matrices and tensors.
- Use matrices and transposes to represent linear maps and compose model stages.
- Solve exact-fitting equations by Gaussian elimination.
- Interpret row-echelon and reduced row-echelon forms, then use pivots to classify solution sets.
- Recognise and compute inverse matrices.

Matrices organise datasets

- ML systems rarely process only one datapoint.
- We often stack many datapoints into one rectangular table.
- Rows represent datapoints; columns represent features.
- Matrix gives us one language for storing data, transforming vectors and composing models.

Example 1. Seven songs are described by danceability, energy and acoustic score.

$$\mathbf{X} = \begin{pmatrix} 0.91 & 0.08 & 0.37 \\ 0.71 & 0.10 & 0.37 \\ 0.25 & 0.20 & 0.46 \\ 0.31 & 0.57 & 0.29 \\ 0.11 & 0.59 & 0.34 \\ 0.12 & 0.57 & 0.35 \\ 0.25 & 0.31 & 0.10 \end{pmatrix} \in \mathbb{R}^{7\times 3}.$$

Each row is one song; each column is one feature.

Matrix. An $m \times d$ matrix is a rectangular array

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1d} \\ a_{21} & a_{22} & \cdots & a_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{md} \end{pmatrix} \in \mathbb{R}^{m \times d}.$$

The entry a_{ij} lies in row i and column j.

Matrices of the same shape are added and scaled entry by entry.

Matrices implement linear maps

- ML tools repeatedly turn one feature vector into another.
- Examples include linear regression, neural-network layers and feature-processing pipelines.
- Linear maps are basic building blocks for these tools.
- We first specify the desired map, then record its coefficients in a matrix.

Linear map. A function $T : \mathbb{R}^d \to \mathbb{R}^m$ is a linear map if, for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$ and $s, t \in \mathbb{R}$,

$$T(s\mathbf{x} + t\mathbf{y}) = sT(\mathbf{x}) + tT(\mathbf{y}).$$

Example 2. For $\mathbf{x} = (x_1, x_2, x_3, x_4, x_5)^{\mathsf{T}}$, suppose we want

$$T(\mathbf{x}) = \begin{pmatrix} (x_1 + x_3 + x_4)/3 \\ 5x_2 - 3x_3 - 12x_5 \\ 2(x_3 + x_4) \end{pmatrix}.$$

Its coefficients are recorded in

$$\mathbf{A} = \begin{pmatrix} \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & 5 & -3 & 0 & -12 \\ 0 & 0 & 2 & 2 & 0 \end{pmatrix}, \qquad T(\mathbf{x}) = \mathbf{A}\mathbf{x}.$$

Matrix–vector product. Let $\mathbf{A} \in \mathbb{R}^{m \times d}$ and $\mathbf{x} \in \mathbb{R}^{d}$. The vector $\mathbf{y} = \mathbf{A}\mathbf{x} \in \mathbb{R}^{m}$ has entries

$$y_i = \sum_{j=1}^{d} a_{ij} x_j.$$

Each output entry is a weighted sum of the input entries.

The shape calculation is

$$(m \times d)(d \times 1) = (m \times 1).$$

Example 2 (cont'd). Apply the matrix $\mathbf{A}$ to $\mathbf{x}$:

$$\mathbf{A} = \begin{pmatrix} \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & 5 & -3 & 0 & -12 \\ 0 & 0 & 2 & 2 & 0 \end{pmatrix} \qquad \mathbf{x} = \begin{pmatrix} 3 \\ 2 \\ -6 \\ 9 \\ -1 \end{pmatrix}.$$

Then

$$\mathbf{Ax} = \begin{pmatrix} (3-6+9)/3 \\ 5(2)-3(-6)-12(-1) \\ 2(-6+9) \end{pmatrix} = \begin{pmatrix} 2 \\ 40 \\ 6 \end{pmatrix}.$$

Proposition 1. Every matrix $\mathbf{A} \in \mathbb{R}^{m \times d}$ defines a linear map $T_{\mathbf{A}} : \mathbb{R}^d \to \mathbb{R}^m$ by

$$T_{\mathbf{A}}(\mathbf{x}) = \mathbf{A}\mathbf{x}.$$

Equivalently, for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$ and $s, t \in \mathbb{R}$,

$$\mathbf{A}(s\mathbf{x} + t\mathbf{y}) = s\mathbf{A}\mathbf{x} + t\mathbf{A}\mathbf{y}.$$

Matrix multiplication composes models

Example 3. Consider two stages:

$$T_{\mathbf{B}}\colon \mathbb{R}^3 \to \mathbb{R}^2, \qquad T_{\mathbf{B}}\left(\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}\right) = \begin{pmatrix} x_1 + x_3 \\ x_2 - x_3 \end{pmatrix},$$

followed by

$$T_{\mathbf{A}}\colon \mathbb{R}^2 \to \mathbb{R}^2, \qquad T_{\mathbf{A}}\left(\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right) = \begin{pmatrix} 2y_1 - y_2 \\ -3y_1 + 4y_2 \end{pmatrix}.$$

Example 3 (cont'd). The maps are represented by

$$\mathbf{B} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}, \qquad \mathbf{A} = \begin{pmatrix} 2 & -1 \\ -3 & 4 \end{pmatrix}.$$

Composing the stages gives

$$\begin{aligned} T_{\mathbf{A}}(T_{\mathbf{B}}(\mathbf{x})) &= T_{\mathbf{A}}\left(\begin{pmatrix} x_1 + x_3 \\ x_2 - x_3 \end{pmatrix}\right) \\ &= \begin{pmatrix} 2(x_1 + x_3) - (x_2 - x_3) \\ -3(x_1 + x_3) + 4(x_2 - x_3) \end{pmatrix} \\ &= \begin{pmatrix} 2x_1 - x_2 + 3x_3 \\ -3x_1 + 4x_2 - 7x_3 \end{pmatrix}. \end{aligned}$$

Matrix product. Let $\mathbf{A} \in \mathbb{R}^{m \times r}$ and $\mathbf{B} \in \mathbb{R}^{r \times d}$. Their product $\mathbf{C} = \mathbf{AB} \in \mathbb{R}^{m \times d}$ has entries

$$c_{ij} = \sum_{q=1}^{r} a_{iq} b_{qj}.$$

Entry (i, j) combines row i of $\mathbf{A}$ with column j of $\mathbf{B}$.

The inner dimensions must agree:

$$(m \times r)(r \times d) = (m \times d).$$

Example 3 (cont'd). Multiply each row of $\mathbf{A}$ by each column of $\mathbf{B}$:

$$\mathbf{A} = \begin{pmatrix} 2 & -1 \\ -3 & 4 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}.$$

$$\mathbf{AB} = \begin{pmatrix} 2(1)+(-1)(0) & 2(0)+(-1)(1) & 2(1)+(-1)(-1) \\ (-3)(1)+4(0) & (-3)(0)+4(1) & (-3)(1)+4(-1) \end{pmatrix}$$

$$= \begin{pmatrix} 2 & -1 & 3 \\ -3 & 4 & -7 \end{pmatrix}.$$

This is precisely the coefficient matrix of the combined map.

- Once the input matrices are fixed, each entry of their product can be computed independently.
- A typical personal-computer CPU has roughly 4–32 powerful cores.
- A modern GPU can have thousands of simpler processing units working simultaneously.
- Large matrix products therefore match the parallel hardware used in deep learning.

Proposition 2. If $\mathbf{A} \in \mathbb{R}^{m \times r}$ and $\mathbf{B} \in \mathbb{R}^{r \times d}$, then, for every $\mathbf{x} \in \mathbb{R}^d$,

$$\mathbf{A}(\mathbf{B}\mathbf{x}) = (\mathbf{A}\mathbf{B})\mathbf{x}.$$

Thus $\mathbf{AB}$ represents the linear map from $\mathbb{R}^d$ to $\mathbb{R}^m$, obtained by applying $\mathbf{B}$ first and $\mathbf{A}$ second.

Proof sketch.

Properties of matrix multiplication

Example 4. Matrix multiplication is not commutative. Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Then

$$\mathbf{AB} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \mathbf{BA} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Hence $\mathbf{AB} \neq \mathbf{BA}$.

Matrix multiplication laws. Whenever the expressions are defined,

$$(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC}),$$

$$\mathbf{A}(\mathbf{B}+\mathbf{C}) = \mathbf{AB}+\mathbf{AC}, \qquad (\mathbf{A}+\mathbf{B})\mathbf{C} = \mathbf{AC}+\mathbf{BC}.$$

Matrix multiplication is associative and distributive, but not commutative.

Transpose and its properties

Transpose. For $\mathbf{A} \in \mathbb{R}^{m \times d}$, its transpose $\mathbf{A}^\mathsf{T} \in \mathbb{R}^{d \times m}$ is obtained by turning rows into columns:

$$(\mathbf{A}^\mathsf{T})_{ji} = a_{ij}.$$

Transposing twice returns the original matrix:

$$(\mathbf{A}^\mathsf{T})^\mathsf{T} = \mathbf{A}.$$

Example 5. For the seven-song data matrix,

$$\mathbf{X}^{\mathsf{T}} = \begin{pmatrix} 0.91 & 0.71 & 0.25 & 0.31 & 0.11 & 0.12 & 0.25 \\ 0.08 & 0.10 & 0.20 & 0.57 & 0.59 & 0.57 & 0.31 \\ 0.37 & 0.37 & 0.46 & 0.29 & 0.34 & 0.35 & 0.10 \end{pmatrix} \in \mathbb{R}^{3\times 7}.$$

Rows of $\mathbf{X}$ become columns of $\mathbf{X}^{\mathsf{T}}$.

Example 6. Let

$$\mathbf{x} = \begin{pmatrix} 3 \\ -4 \\ 12 \end{pmatrix}.$$

Then

$$\mathbf{x}^\mathsf{T}\mathbf{x} = \begin{pmatrix} 3 & -4 & 12 \end{pmatrix} \begin{pmatrix} 3 \\ -4 \\ 12 \end{pmatrix}$$

$$= 3^2 + (-4)^2 + 12^2 = 169 = \|\mathbf{x}\|_2^2.$$

More generally, $\|\mathbf{x}\|_2^2 = \mathbf{x}^\mathsf{T}\mathbf{x}$ for every $\mathbf{x} \in \mathbb{R}^d$.

Transpose rules. For matrices of compatible shapes and $s \in \mathbb{R}$,

$$(\mathbf{A} + \mathbf{B})^\mathsf{T} = \mathbf{A}^\mathsf{T} + \mathbf{B}^\mathsf{T}, \qquad (s\mathbf{A})^\mathsf{T} = s\mathbf{A}^\mathsf{T},$$

and

$$(\mathbf{AB})^\mathsf{T} = \mathbf{B}^\mathsf{T}\mathbf{A}^\mathsf{T}.$$

The order reverses when a product is transposed.

Linear-regression loss. Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ be a data matrix, let $\mathbf{y} \in \mathbb{R}^n$ contain the targets and let $\mathbf{w} \in \mathbb{R}^d$ be the weight vector. The predictions are

$$\widehat{\mathbf{y}} = \mathbf{X}\mathbf{w}.$$

The mean squared error can be written as

$$\begin{aligned} L(\mathbf{w}) = \frac{1}{n}\sum_{i=1}^{n}(\mathbf{x}_i^\mathsf{T}\mathbf{w} - y_i)^2 &= \frac{1}{n}\|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 \\ &= \frac{1}{n}(\mathbf{X}\mathbf{w} - \mathbf{y})^\mathsf{T}(\mathbf{X}\mathbf{w} - \mathbf{y}). \end{aligned}$$

Linear-regression loss (cont'd). Using transpose rules,

$$\begin{aligned} nL(\mathbf{w}) &= (\mathbf{X}\mathbf{w} - \mathbf{y})^\mathsf{T}(\mathbf{X}\mathbf{w} - \mathbf{y}) \\ &= (\mathbf{w}^\mathsf{T}\mathbf{X}^\mathsf{T} - \mathbf{y}^\mathsf{T})(\mathbf{X}\mathbf{w} - \mathbf{y}) \\ &= \mathbf{w}^\mathsf{T}\mathbf{X}^\mathsf{T}\mathbf{X}\mathbf{w} - \mathbf{y}^\mathsf{T}\mathbf{X}\mathbf{w} - \mathbf{w}^\mathsf{T}\mathbf{X}^\mathsf{T}\mathbf{y} + \mathbf{y}^\mathsf{T}\mathbf{y}. \end{aligned}$$

The two cross terms are equal because they are scalars:

$$\mathbf{w}^\mathsf{T}\mathbf{X}^\mathsf{T}\mathbf{y} = \mathbf{y}^\mathsf{T}\mathbf{X}\mathbf{w}.$$

Since $\mathbf{y}^\mathsf{T}\mathbf{y}$ is independent of $\mathbf{w}$, minimizing $L(\mathbf{w})$ is equivalent to minimizing $\mathbf{w}^\mathsf{T}\mathbf{X}^\mathsf{T}\mathbf{X}\mathbf{w} - 2\mathbf{y}^\mathsf{T}\mathbf{X}\mathbf{w}$.

Batch transformations and layered models

Batch transformation. Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ store $\mathbf{x}_i^\mathsf{T}$ in row i, and let $\mathbf{A} \in \mathbb{R}^{p \times d}$ transform one datapoint. The transformed datapoint is $\mathbf{A}\mathbf{x}_i$, whose row representation is

$$(\mathbf{A}\mathbf{x}_i)^\mathsf{T} = \mathbf{x}_i^\mathsf{T}\mathbf{A}^\mathsf{T}.$$

Stacking these rows gives

$$\mathbf{Y} = \mathbf{X}\mathbf{A}^\mathsf{T} \in \mathbb{R}^{n \times p}.$$

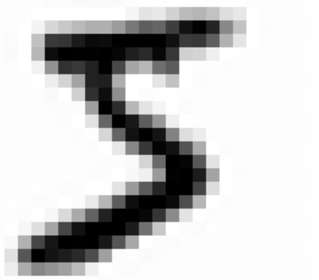
Two-layer linear network. Suppose

$$\mathbf{h} = \mathbf{B}\mathbf{x}, \qquad \mathbf{y} = \mathbf{A}\mathbf{h} = (\mathbf{A}\mathbf{B})\mathbf{x}.$$

Thus two consecutive linear layers are equivalent to one linear map. For a batch $\mathbf{X} \in \mathbb{R}^{n \times d}$, the transformed dataset is

$$\mathbf{X}(\mathbf{A}\mathbf{B})^{\mathsf{T}} = \mathbf{X}\mathbf{B}^{\mathsf{T}}\mathbf{A}^{\mathsf{T}} \in \mathbb{R}^{n \times p}.$$

- So far, each datapoint is represented by a vector.
- Image datapoints(for computer vision tasks) are better represented by matrices to preserve geometric information.



- In this case, we use tensors, a natural generalisation of matrices. For example, n MNIST images are represented by a single tensor in $\mathbb{R}^{n\times 28\times 28}$ in PyTorch.
- Read Lecture Note 2 for more details and examples.

Exact model fitting becomes a linear system

- In supervised learning, datapoint $\mathbf{x}_i \in \mathbb{R}^d$ comes with target y_i.
- A linear model with weights $\mathbf{w} \in \mathbb{R}^d$ predicts $\mathbf{x}_i^\mathsf{T}\mathbf{w}$.
- Exact fitting requires $\mathbf{x}_i^\mathsf{T}\mathbf{w} = y_i$ for every training datapoint.
- Stacking the requirements gives the linear system $\mathbf{X}\mathbf{w} = \mathbf{y}$.

Example 7. Three labelled datapoints impose exact requirements:

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix},\ y_1 = 5, \quad \mathbf{x}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix},\ y_2 = 4, \quad \mathbf{x}_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix},\ y_3 = 3.$$

For $f_{\mathbf{w}}(\mathbf{x}) = \mathbf{w}^{\mathsf{T}}\mathbf{x}$, what weights fit all three datapoints?

Example 7 (cont'd). The requirements are

$$w_1 + w_2 = 5, \qquad w_2 + w_3 = 4, \qquad w_1 + w_3 = 3.$$

Equivalently,

$$\underbrace{\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}}_{\mathbf{A}} \underbrace{\begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix}}_{\mathbf{w}} = \underbrace{\begin{pmatrix} 5 \\ 4 \\ 3 \end{pmatrix}}_{\mathbf{b}}.$$

The solution is $\mathbf{w} = (2, 3, 1)^{\mathsf{T}}$.

Linear system. Let $\mathbf{A} \in \mathbb{R}^{n \times d}$ and $\mathbf{b} \in \mathbb{R}^n$. A linear system with unknown $\mathbf{w} \in \mathbb{R}^d$ is

$$\mathbf{A}\mathbf{w} = \mathbf{b}.$$

A solution satisfies all n equations.

The system is consistent if it has at least one solution, and inconsistent if it has none.

Augmented matrix. We adjoin the required outputs to the coefficients:

$$[\mathbf{A} \mid \mathbf{b}] = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1d} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2d} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nd} & b_n \end{array}\right).$$

The vertical bar separates coefficients from outputs.

- With three unknowns, a nontrivial linear equation describes a plane.
- Several planes may meet at one point, share a line or have no common point.
- These cases correspond to one solution, infinitely many solutions or no solution.
- Gaussian elimination detects all three cases without requiring a picture.

Gaussian elimination preserves solution set

Example 8. Begin with the augmented matrix from Example 7:

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 5 \\ 0 & 1 & 1 & 4 \\ 1 & 0 & 1 & 3 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 - R_1} \left(\begin{array}{ccc|c} 1 & 1 & 0 & 5 \\ 0 & 1 & 1 & 4 \\ 0 & -1 & 1 & -2 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{ccc|c} 1 & 1 & 0 & 5 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 2 & 2 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow \frac{1}{2} R_3} \left(\begin{array}{ccc|c} 1 & 1 & 0 & 5 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 1 & 1 \end{array}\right).$$

Example 8 (cont'd).

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 5 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 1 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - R_3} \left(\begin{array}{ccc|c} 1 & 1 & 0 & 5 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1 \end{array}\right)$$

$$\xrightarrow{R_1 \leftarrow R_1 - R_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1 \end{array}\right).$$

The final matrix states $w_1 = 2$, $w_2 = 3$ and $w_3 = 1$.

Elementary row operations.

1. Interchange two rows.
2. Multiply a row by a nonzero scalar.
3. Add a scalar multiple of one row to another row.

Two matrices are row equivalent if a sequence of these operations transforms one into the other.

Proposition 3. Row-equivalent augmented matrices represent systems with the same solution set.

Every elementary row operation is reversible:

- swap the rows back;
- divide by the nonzero scalar;
- subtract the same multiple of the other row.

Row-echelon form. In each nonzero row, the first nonzero entry is its pivot. A matrix is in row-echelon form if:

1. all zero rows lie below all nonzero rows;
2. each lower pivot lies strictly to the right of the pivot above it;
3. every entry below a pivot is zero.

The pivots form a staircase.

Reduced row-echelon form. In addition to being in echelon form:

- every pivot equals 1;
- each pivot is the only nonzero entry in its column.

A coefficient column with a pivot gives a pivot variable.

A coefficient column without a pivot gives a free variable.

Example 9. Consider the augmented matrix

$$\mathbf{E} = \left(\begin{array}{cccc|c} 3 & 6 & -18 & 9 & 87 \\ 0 & 5 & -25 & -4 & 11 \\ 0 & 0 & 0 & 7 & 42 \end{array}\right).$$

It is in row-echelon form: the pivots lie in columns 1, 2 and 4, move strictly to the right and have only zeros below them.

It is not in reduced row-echelon form: its pivots are not 1, and their columns contain nonzero entries above them.

Example 9 (cont'd). Starting from

$$\left(\begin{array}{cccc|c} 3 & 6 & -18 & 9 & 87 \\ 0 & 5 & -25 & -4 & 11 \\ 0 & 0 & 0 & 7 & 42 \end{array}\right) \xrightarrow{R_3 \leftarrow \frac{1}{7} R_3}$$

$$\xrightarrow{R_2 \leftarrow R_2 + 4R_3}$$

Example 9 (cont'd).

$$\xrightarrow{R_1 \leftarrow R_1 - 9R_3}$$

$$\xrightarrow{R_2 \leftarrow \frac{1}{5}R_2}$$

Example 9 (cont'd).

$$\xrightarrow{R_1 \leftarrow R_1 - 6R_2}$$

$$\xrightarrow{R_1 \leftarrow \frac{1}{3}R_1}$$

Gaussian elimination.

1. Work from left to right through the columns.
2. Move a nonzero entry into the next pivot position.
3. Eliminate every entry below that pivot.
4. Continue to obtain row-echelon form, then back-substitute.

Gauss–Jordan elimination also scales pivots to 1 and eliminates above them.

Pivots reveal all possible outcomes

Example 10. In Example 8, the original system reduces to

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1 \end{array}\right).$$

Every coefficient column contains a pivot, so the system has the unique solution

$$(w_1, w_2, w_3) = (2, 3, 1).$$

Example 10 (cont'd). Replace the third equation by $w_1 - w_3 = 1$:

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 5 \\ 0 & 1 & 1 & 4 \\ 1 & 0 & -1 & 1 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 1 & 0 & 5 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

The variable w_3 is free. Setting $w_3 = t$ gives

$$w_2 = 4 - t, \qquad w_1 = 1 + t.$$

There are infinitely many solutions.

Example 10 (cont'd). Change the new third target from 1 to 2:

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 5 \\ 0 & 1 & 1 & 4 \\ 1 & 0 & -1 & 2 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 1 & 0 & 5 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{array}\right).$$

The final row asserts

$$0 = 1.$$

The system is inconsistent.

Theorem 4. Let $[\mathbf{A} \mid \mathbf{b}]$ be in row-echelon form.

- The system is inconsistent exactly when it contains a row $\left(\begin{array}{ccc|c} 0 & \cdots & 0 & c \end{array} \right)$ with $c \neq 0$.
- If consistent, the solution is unique exactly when every coefficient column contains a pivot.
- Otherwise, at least one variable is free and there are infinitely many solutions.

Example 11. A model has six weights and four exact requirements:

$$\left(\begin{array}{cccccc|c} 0 & 2 & 0 & 0 & 0 & 4 & 6 \\ 1 & 0 & 0 & 0 & 2 & 0 & 2 \\ 0 & 0 & -3 & 0 & -9 & -6 & -12 \\ 1 & 0 & 0 & 1 & 5 & 4 & 3 \end{array}\right).$$

First interchange R_1 and R_2.

Example 11 (cont'd). Next scale rows and perform a row replacement:

$$R_2 \leftarrow \tfrac{1}{2}R_2, \qquad R_3 \leftarrow -\tfrac{1}{3}R_3, \qquad R_4 \leftarrow R_4 - R_1.$$

This gives

$$\left(\begin{array}{cccccc|c} 1 & 0 & 0 & 0 & 2 & 0 & 2 \\ 0 & 1 & 0 & 0 & 0 & 2 & 3 \\ 0 & 0 & 1 & 0 & 3 & 2 & 4 \\ 0 & 0 & 0 & 1 & 3 & 4 & 1 \end{array}\right).$$

Example 11 (cont'd). The pivot variables are w_1, w_2, w_3, w_4; the free variables are w_5, w_6.

Setting $w_5 = s$ and $w_6 = t$ gives

$$\mathbf{w} = \begin{pmatrix} 2 - 2s \\ 3 - 2t \\ 4 - 3s - 2t \\ 1 - 3s - 4t \\ s \\ t \end{pmatrix}, \qquad s, t \in \mathbb{R}.$$

Inverse matrices undo square transformations

Example 12. Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \qquad \mathbf{B} = \frac{1}{2}\begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{pmatrix}.$$

Direct multiplication gives

$$\mathbf{AB} = \mathbf{BA} = \mathbf{I}_3.$$

Thus $\mathbf{B}$ undoes the transformation represented by $\mathbf{A}$.

Inverse matrix. The identity matrix satisfies

$$\mathbf{I}_d\mathbf{x} = \mathbf{x}.$$

A square matrix $\mathbf{A} \in \mathbb{R}^{d\times d}$ is invertible if there is a matrix $\mathbf{A}^{-1}$ such that

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}_d.$$

The matrix $\mathbf{A}^{-1}$ is the inverse of $\mathbf{A}$.

Proposition 5. If $\mathbf{A} \in \mathbb{R}^{d \times d}$ is invertible, then for every $\mathbf{b} \in \mathbb{R}^d$, the system $\mathbf{A}\mathbf{w} = \mathbf{b}$ has the unique solution

$$\mathbf{w} = \mathbf{A}^{-1}\mathbf{b}.$$

- Gauss–Jordan elimination computes $\mathbf{A}^{-1}$ by solving d systems at once.
- Begin with the block matrix $[\mathbf{A} \mid \mathbf{I}_d]$.
- Apply every row operation to both blocks.
- If the result is $[\mathbf{I}_d \mid \mathbf{B}]$, then $\mathbf{B} = \mathbf{A}^{-1}$.
- If the left block has fewer than d pivots, $\mathbf{A}$ is not invertible.

Example 13. Start with

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right).$$

Apply

$$R_3 \leftarrow R_3 - R_1, \qquad R_3 \leftarrow R_3 + R_2, \qquad R_3 \leftarrow \tfrac{1}{2} R_3.$$

Example 13 (cont'd). Continue with

$$R_2 \leftarrow R_2 - R_3, \qquad R_1 \leftarrow R_1 - R_2.$$

The result is

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array}\right).$$

Therefore the right block is $\mathbf{A}^{-1}$.

Example 14. Consider

$$\mathbf{A} = \begin{pmatrix} 4 & 7 & -3 & -6 \\ 0 & 5 & 8 & 3 \\ 0 & 0 & 6 & 6 \\ 8 & 29 & -6 & -27 \end{pmatrix}.$$

Gaussian elimination gives a left block with echelon form

$$\begin{pmatrix} 4 & 7 & -3 & -6 \\ 0 & 5 & 8 & 3 \\ 0 & 0 & 6 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

There are only three pivots, so $\mathbf{A}$ is not invertible.

Takeaways

- Matrices organise datasets and represent linear maps.
- Matrix multiplication composes model stages and enables parallel computation.
- Row operations preserve solutions to linear systems.
- Pivots distinguish unique, infinite and impossible solution sets.
- Inverse matrices undo invertible square transformations.